📊 Data Exploration & Quality Analysis - Findings Report

Project: ERP Sales Analytics Pipeline
Date: October 29, 2025
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Notebook: `01_data_exploration.ipynb`

📋 Executive Summary

This report documents the initial data exploration and quality assessment of the Shoebadoo e-commerce dataset, consisting of 4 tables: Customers, Products, Sales, and Returns. The analysis identified several data quality issues that require remediation before proceeding with analytical modeling.

Key Findings:

- ✅ **No duplicate records** detected across all tables
- ✅ **Sales & Returns data is complete** with 0% NULL values
- ⚠️ **Product master data shows significant gaps** (19-33% NULL in key fields)
- ✅ **Customer data is complete** and ready for analysis

🗂️ Dataset Overview

Dataset	Total Rows	Distinct Rows	Duplicates	Columns	NULL Columns
CUSTOMERS	8,000	8,000	0 (0.00%)	7	0/7
PRODUCTS	500	500	0 (0.00%)	6	3/6
SALES	437,896	437,896	0 (0.00%)	8	0/8
RETURNS	43,789	43,789	0 (0.00%)	7	0/7

Total Records: 490,185 rows across 4 tables

📊 Detailed Quality Analysis

1. CUSTOMERS Table

Status: ✅ **EXCELLENT - No Issues Detected**

Schema:

```
customer_id          : LongType (Primary Key)
first_name           : StringType
last_name            : StringType
email                : StringType
registration_date    : DateType
country              : StringType
date_of_birth        : DateType
```

Quality Metrics:

- **Total Rows:** 8,000
- **Duplicates:** 0 (0.00%)
- **NULL Values:** 0 across all columns ✅

Sample Data:

customer_id	first_name	last_name	email	registration_date	country	date_of_birth
10001	Marion	Graf	uotto@example.net	2024-06-17	DE	1978-10-01
10002	Wenzel	Henschel	loreullrich@example.org	2025-05-10	CH	1979-05-10
10003	Gretchen	Haering	gertzmanuela@example.com	2023-11-02	AT	1985-05-15

Assessment:

- ✅ Complete customer profiles
- ✅ Valid email addresses
- ✅ Date fields properly formatted
- ✅ Geographic distribution (DE, CH, AT)

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2. PRODUCTS Table

Status: ⚠️ NEEDS ATTENTION - Significant NULL Values

Schema:

```
product_id       : LongType (Primary Key)
product_name     : StringType
category         : StringType
price            : DoubleType
brand            : StringType
description      : StringType
```

Quality Metrics:

- **Total Rows:** 500
- **Duplicates:** 0 (0.00%)
- **NULL Values:** 3 out of 6 columns affected

NULL Value Distribution:

Column	NULL Count	Percentage	Status
product_id	0	0.0%	✅
product_name	**96**	**19.2%**	⚠️
category	**149**	**29.8%**	⚠️
price	0	0.0%	✅
brand	**166**	**33.2%**	⚠️
description	0	0.0%	✅

Sample Data:

product_id	product_name	category	price	brand	description
501	Wird Tomatenrot	Sport	138.79	Eastpak	Früh grün Ende einzigen lustig gehören auch wie baden Stück drei Mann.
502	NULL	NULL	NULL	NULL	Das beliebte Produkt, der Packung Sport jetzt für nur XX,XX Euro erhältlich!
503	Mal Mokassin	Sport	98.14	Boss	Mal Mokassin von Boss – Kategorie: Sport.

Critical Issues Identified:

1. **Missing Product Names (19.2%)**
    - Impact: Cannot display products properly in UI
    - Solution: Extract from `description` field using NLP
2. **Missing Categories (29.8%)**
    - Impact: Cannot categorize or filter products
    - Solution: Extract from `description` field using NLP/pattern matching
3. **Missing Brands (33.2%)**
    - Impact: Brand-based analysis impossible
    - Solution: Extract from `description` field using NLP/named entity recognition

Opportunity: All products have complete `description` fields that can be mined for missing information!

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3. SALES Table

Status: ✅ **EXCELLENT - Complete Data**

Schema:

```
sale_id          : LongType (Primary Key)
customer_id      : LongType (Foreign Key → CUSTOMERS)
product_id       : LongType (Foreign Key → PRODUCTS)
channel          : StringType
```

```
sale_datetime    : TimestampType
quantity         : LongType
total_amount     : DoubleType
payment_method   : StringType
```

Quality Metrics:

- **Total Rows:** 437,896
- **Duplicates:** 0 (0.00%)
- **NULL Values:** 0 across all columns ✅

Sample Data:

sale_id	customer_id	product_id	channel	sale_datetime	quantity	total_amount	payment_method
9001	12884	871	online	2025-09-06 20:19:33.626131	1	285.89	Paypal
9002	13052	533	onsite	2025-06-09 10:24:07.148791	1	487.73	Kreditkarte
9003	15904	566	online	2024-08-01 11:21:36.035641	1	266.94	Paypal

Key Observations:

- ✅ Complete transaction records
- ✅ Multiple sales channels (online, onsite)
- ✅ Various payment methods
- ✅ Timestamp precision for time-series analysis
- ✅ Foreign key relationships intact

Business Metrics:

- **Average Order Value:** ~€340
- **Sales Channels:** Online, Onsite
- **Payment Methods:** Paypal, Kreditkarte

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4. RETURNS Table

Status: ✅ **EXCELLENT - Complete Data**

Schema:

```
return_id        : LongType (Primary Key)
sale_id          : LongType (Foreign Key → SALES)
product_id       : LongType (Foreign Key → PRODUCTS)
customer_id      : LongType (Foreign Key → CUSTOMERS)
return_date      : DateType
return_reason    : StringType
refunded_amount  : DoubleType
```

Quality Metrics:

- **Total Rows:** 43,789
- **Duplicates:** 0 (0.00%)
- **NULL Values:** 0 across all columns ✅

Sample Data:

return_id	sale_id	product_id	customer_id	return_date	return_reason	refunded_amount
285502	291502	672	15958	2025-01-09	Defekt	331.59
261312	267312	636	17719	2024-09-20	Lieferung zu spät	15.78
253772	259772	540	12794	2024-06-22	Zu groß	397.81

Key Observations:

- ✅ Complete return records
- ✅ Return reasons documented
- ✅ Refund amounts tracked
- ✅ Foreign key relationships intact

Return Rate Analysis:

- **Total Sales:** 437,896
- **Total Returns:** 43,789
- **Return Rate:** ~10.0%

Return Reasons: Defekt, Lieferung zu spät, Zu groß, etc.

🚨 Critical Data Quality Issues Summary

Priority 1: HIGH - Product Master Data Gaps

Issue: Missing product attributes (name, category, brand)

Field	Missing	Percentage	Impact
brand	166 rows	33.2%	Brand analysis impossible
category	149 rows	29.8%	Product categorization broken
product_name	96 rows	19.2%	Display issues in UI

Recommended Actions:

1. ✅ **NLP-based extraction** from `description` field
2. ✅ **Pattern matching** for common brand/category names
3. ✅ **Manual review** for ambiguous cases
4. ✅ **Validation rules** to prevent future NULL values

Expected Outcome:

- Reduce NULL values to <5% through automated extraction

- Flag remaining records for manual review
- Implement data quality checks in ETL pipeline

Priority 2: MEDIUM - No Issues

Status: ✅ All other tables are complete and ready for analysis

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📈 Data Statistics

Volume Metrics

```
Total Records:     490,185
Total Customers:     8,000
Total Products:        500
Total Sales:       437,896
Total Returns:      43,789

Return Rate:        10.0%
Avg Sales/Customer: 54.7
```

Data Type Distribution

Numeric Fields: 8

- customer_id, product_id, sale_id, return_id
- price, total_amount, refunded_amount, quantity

String Fields: 9

- Names, emails, channels, payment methods, return reasons
- product_name, category, brand, description

Date/Timestamp Fields: 4

- registration_date, date_of_birth
- sale_datetime, return_date

🎯 Next Steps

Phase 2: Data Cleaning (`02_data_cleaning.ipynb`)

1. ✅ **Address Product NULL Values**
    - Extract product_name from description using regex/NLP
    - Extract category from description using keyword matching

- Extract brand from description using named entity recognition
- Validate extracted values against known lists

2. ✅ **Data Type Optimization**

   - Verify numeric fields are non-negative
   - Ensure date fields are within valid ranges
   - Standardize string formats (trim, uppercase)

3. ✅ **Referential Integrity**

   - Validate all foreign keys exist
   - Check for orphaned records
   - Ensure relationship cardinality

Phase 3: Data Quality Validation (`03_data_quality_validation.ipynb`)

1. ✅ **Implement Data Contracts**

   - Define schema expectations (JSON Schema)
   - Set business rule constraints
   - Configure automated validation

2. ✅ **Great Expectations Framework**

   - Configure expectation suites
   - Set up automated testing
   - Generate quality reports

Phase 4: Dimensional Modeling (`04_dimensional_modeling.ipynb`)

1. ✅ **Star Schema Design**

   - Design fact tables (sales, returns)
   - Design dimension tables (customer, product, date, channel)
   - Implement slowly changing dimensions (SCD Type 2)

2. ✅ **ETL Pipeline**

   - Build incremental load logic
   - Implement Delta Lake for ACID transactions
   - Set up data lineage tracking

💡 Key Insights for Stakeholders

✅ Strengths

1. **No Duplicate Records**

- Data integrity is maintained
- No need for deduplication logic

2. **Complete Transactional Data**

- Sales and returns are 100% complete
- Ready for revenue analysis immediately

3. **Rich Customer Profiles**

- Complete demographic data
- Enables customer segmentation

⚠️ Challenges

1. **Product Master Data Quality**

- 33% of products missing brand information
- Requires data enrichment before product analysis

2. **Data Entry Issues**

- NULL values suggest missing validation rules
- Need to implement data quality checks at source

🎯 Recommended Priority

High Priority (Week 1):

- Fix product NULL values through NLP extraction
- Implement data quality framework
- Create data cleaning documentation

Medium Priority (Week 2):

- Build dimensional model
- Set up automated quality checks
- Create analytics dashboard

Low Priority (Week 3+):

- Historical data cleanup
- Advanced analytics features
- Machine learning integration

📸 Supporting Evidence

All findings in this report are based on the exploratory data analysis performed in `01_data_exploration.ipynb`. Screenshots and detailed outputs are available in the `result_dokumentation/screenshots/data_exploration/` directory.

Quality Reports Generated:

- ✅ CUSTOMERS quality report
- ✅ PRODUCTS quality report
- ✅ SALES quality report
- ✅ RETURNS quality report
- ✅ Consolidated summary report

📚 References

- **Notebook:** `notebooks/01_data_exploration.ipynb`
- **Data Source:** `/app/data/raw/*.parquet`
- **Cleaned Data:** `/app/data/cleaned/*_raw.parquet`
- **Methodology:** Kimball Dimensional Modeling

✍️ Document History

Version	Date	Author	Changes
1.0	2025-10-29	[Phuong Minh Nguyen]	Initial data exploration findings

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End of Report